

ZIMO ZHANG

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EDUCATION BACKGROUND

University of Toronto (U of T)

09/2025-05/2027 (estimated)

Master of Engineering in Electrical & Computer Engineering

Toronto, ON

Specialization: Software Engineering

University of British Columbia (UBC)

09/2021-05/2025

Bachelor of Science in Mathematic with Minor in Data Science

Vancouver, BC

GPA: 3.3/4.33

Core Courses: Intro to Data Science, Intro to Linear Programming, Visualization for Data Science, Intro to Mathematical Modelling, Applied Machine Learning, Basic Algorithms and Data Structures, Applied Linear Algebra, Calculus, Statistical Modelling for Data Science, Intro to Relational Databases, Intro to Artificial Intelligence, Elementary Differential Equations II, Software Construction, Matrix Algebra, Elementary Statistics for Applications, Computation, Programs, and Programming, Statistical Inference for Data Science, Intro to Probability

PROFESSIONAL SKILLS

- C++ (2 yr), Java (2 yr), Python (3 yrs), R (3 yrs)
- SOA Probability

PUBLICATION

Zimo Z. (2024). A Study of Ocular Disease Prediction Based on Convolutional Neural Network Algorithm, 2024 6th International Conference on Computer Science and Intelligent Communication (CSIC 2024). (Accepted)

RESEARCH EXPERIENCE

Artificial Intelligence and Machine Learning- Research on Machine Learning Theories such as Neural Networks and Data Dimensionality Reduction, and Practical Application of Python Programming

07/2024-09/2024

Participant | Prof. Mark Vogelsberger

- Participated in the online research seminar and learned about mathematical concepts, fundamental machine learning algorithms
- Implemented machine learning algorithms like SVM, linear regression, logistic regression, decision trees, recurrent neural networks, and graphical user interfaces to deal with the practical programming problems
- Developed a disease prediction model based on fundus images using a Convolutional Neural Networks (CNN) algorithm
- Completed the paper of A Study of Ocular Disease Prediction Based on Convolutional Neural Network Algorithm

INTERNSHIP

Shaanxi Zilu Electronic Technology Co.

07/2024- 08/2024

Software Development Intern

- Assisted the project manager in collecting and categorizing customer requirements in the Mobile Nursing Software project, and carried out the task of entering and perfecting basic data
- Collaborated with the team from the initial planning and design to the later debugging and optimization, meticulously arranging the network architecture to ensure smooth information flow
- Participated in requirement investigations, ranging from patient information management to medical resource scheduling, from

clinical decision support to medical data analysis

- Helped medical staff quickly adapt to the new system, ensuring seamless integration of the system with actual medical business

Chinese Institute for Brain Research

06/2025- 08/2025

Algorithm Engineer Intern, Computing and Data Science Core

- Performed data annotation for mouse epileptic seizures and executed data segmentation and deep-learning model training for epileptic seizure prediction tasks.
- Demonstrated diligence and strong sense of responsibility in all assigned tasks, strictly adhering to institutional regulations and maintaining effective collaboration with the research team.
- Proposed original ideas and constructive solutions, actively exploring diverse domains and applying innovative thinking to cross-disciplinary problems.
- Adapted quickly to new methodologies and tools, ensuring high-quality outcomes while broadening knowledge across computing, neuroscience, and data science.

WORK EXPERIENCE

Haidilao Hotpot – West Broadway, Vancouver

11/2024- 05/2025

Part-time Server

- Greeted customers and gave a brief menu introduction
- Took orders, served dishes, and refilled soup
- Assisted with take-out and delivery orders to ensure accuracy and timeliness
- Monitored private dining rooms, maintained cleanliness, and supported guests' additional requests

Ben Gong's Tea – Vancouver

07/2023- 08/2023

Part-time Barista

- Prepared a variety of bubble tea drinks and desserts according to standard recipes
- Handled customer orders and payments efficiently in a fast-paced environment
- Maintained cleanliness of the workstation and ensured consistent product quality

COURSEWORK

Data Visualization Analysis on Smoking and Cholesterol

01/2025- 04/2025

- Analyzed the relationship between smoking status and cholesterol levels using the Korean National Health and Nutrition Examination Survey (KNHANES) dataset
- Developed 3 interactive visualizations with Python Altair: Compared cholesterol distribution among non-smokers, former smokers, and current smokers, investigated age and gender effects on smoking-cholesterol relationships, explored correlations between BMI, blood pressure and cholesterol in smoking populations
- Demonstrated cardiovascular impacts of smoking through interactive visualizations to support health policy formulation

Travel Manager Application

01/2024- 04/2024

- Conducted the requirement analysis of the application to understand the needs of travel agencies and their clients, defined the scope and functions of the application, including the management of travel destinations, clients, tours and activities
- Design the schema of the database to store related information, and implemented the database with appropriate tables, relationships, and constraints to ensure data integrity and efficiency
- Developed the application using Java and Oracle database, and created a user-friendly interface for clients in planning their travel

routes and preferences

Predictive Analysis on Prices of Rental Apartments in USA using Major Rental Price Influencers

09/2023-12/2023

- Collaborated with a team in STAT 301 to analyze key factors influencing rental apartment prices in the USA using machine learning techniques
- Worked with a dataset of 10,000 rental apartment listings, collected from 12 online platforms, cleaned and preprocessed the data, addressing missing values and multicollinearity issues
- Employed Python, Pandas, and Matplotlib for data exploration and visualization, identifying key correlations between variables like apartment size, pet policies, and rental prices
- Built and optimized a linear regression model, effectively improved the efficiency and accuracy of the prediction on rental prices

E-binder of Photo Cards Application

07/2023-08/2023

- Developed an e-binder application using Java to manage K-pop photo card collections
- Created a database to store and organize information about the photo cards, ensured efficient data retrieval and storage for user inputs
- Conducted thorough testing to ensure the application runs smoothly, optimizing the application for performance and usability

Heart Disease Prediction

09/2022-12/2022

- Conducted a research on heart disease prediction to understand existing knowledge and identify potential factors influencing heart disease risk
- Performed data processing and analysis and used statistical techniques and visualization tools to identify significant factors related to heart disease
- Built various machine learning models to predict the likelihood of heart disease, ensuring the selected model was accurate in predicting heart disease